FEN EDEBİYAT FAKÜLTESİ
FACULTY OF ARTS AND SCIENCES
FİZİK BÖLÜMÜ
DEPARTMENT OF PHYSICS

Syllabus

PHYS331 – ELECTROMAGNETIC THEORY I
2026 – 2027 FALL SEMESTER

AI Artificial intelligence was used in the preparation of this document.

Basis & Scope

(1) This syllabus has been prepared based on *Middle East Technical University Rules and Regulations Governing Undergraduate Studies* and applies only to the course and the semester in the subtitle.

(2) This document only covers *what* the course rules are; to understand *why*, feel free to contact me.[1] You may also find the following links useful in this regard:
https://soneralbayrak.com/teaching/files/teaching_methodology.pdf
https://soneralbayrak.com/teaching/files/exam_complaints.pdf

Academic Accommodations

(3) If you have official documentation of a disability, you may apply for a *Letter for Academic Accommodations* by contacting *METU Disability Support Office*: please do so if this applies.

(4) Even if you do not have an officially documented disability, contact me if you have any special situation such as *physical disability, learning disability, ADHD, chronic health issue, mental health condition, complications with a pregnancy, temporarily disabling injury or circumstance, upcoming surgery* and so on.

(5) You should contact me as soon as a need for an academic accommodation arises so that I can do my best to tailor your learning experience to your needs: please do not delay or postpone informing me of your situation!

Academic Integrity

(6) The METU Code of Honor and the academic integrity guide, available via the *METU Code of Ethics & Core Values* website, are applicable to this course. For further details, please refer to https://soneralbayrak.com/teaching/files/academic_integrity.pdf.

(7) During examinations, please adhere to the regulations detailed in
https://soneralbayrak.com/teaching/files/exam_rules.pdf.

Help Outside the Lectures

(8) The teaching assistant assigned to the class (if any) will introduce themselves in due time through ODTÜClass/email, and detail the help that they can provide, such as office hours or problem-solving recitations.

(9) The responsibilities of teaching assistants towards the class and teaching in general are detailed here: https://soneralbayrak.com/teaching/files/ta_guide.pdf

[1] My official email: soneralb@metu.edu.tr.

(10) My weekly schedule, my office hours, possibility of appointments, and any other related information are available at https://soneralbayrak.com/schedule.

Attendance

(11) Attendance is mandatory and is checked several times at randomly chosen lectures during the semester, through the ODTÜClass attendance plugin or the old-fashioned pen & paper.[2]

(12) Any student who was absent at four or more attendance checks will be deemed to have insufficient participation.

(13) Any student with insufficient participation who would otherwise receive the letter grade FF/FD will be given NA. Besides this, attendance does not affect the letter grades.

(14) As the attendance only serves to distinguish *academic failure* (the student attended the classes yet failed) from *procedural failure* (the student failed, but it may have been due to a lack of effort), there is no excuse mechanism for attendance.

Midterm and Final Examinations

(15) The midterm examination dates are available in Article **(43)**. The final examination date is announced at the end of the semester.

(16) Exam locations and exam hours are announced via ODTÜClass/email in due time.

(17) The exams are cumulative: you are responsible for everything covered from the start of the semester until the date of the exam.

(18) Each exam consists of several questions, each question with eight choices and an answer box below. Students are expected to select the correct choice ***and*** provide a *step-by-step explanation of how they solved the question* in the answer box.[3]

(19) If no choice is selected, the question is awarded 1/8 of the question's points.[4]

(20) If the correct choice is selected but the explanation in the answer box is insufficient,[5] the question may be awarded 1/8 of the question's points instead of full points.

(21) If an incorrect choice is selected, then the question is awarded zero points, regardless of the content of the answer box.

(22) In the examinations, you will also be provided with this formula sheet (to be updated as deemed necessary): https://soneralbayrak.com/teaching/files/formula_sheet.pdf.

Admissible Excuses for Missing an Examination

(23) You are entitled to an admissible excuse for an exam if all of the following conditions are satisfied:

- **(a)** you have missed that exam (taking an exam and submitting an empty paper does not satisfy this condition);
- **(b)** you have a medical report that covers the exam date;

[2] Students who are present in the lecture but unable to access the ODTÜClass attendance plugin at that moment should inform me immediately after the lecture (before leaving the classroom).

[3] They need not and should not give the full solution in the answer box; instead, they should *explain* the solution in a consistent and self-contained manner.

[4] Leaving a question empty shows an awareness of one's lack of knowledge.

[5] The explanation is supposed to be succinct, well-articulated, and correct both scientifically and mathematically.

(c) you have submitted your medical report to the *Health and Guidance Center* within 5 days after the period covered by the report ends;

(d) you have emailed me a picture of your medical report as soon as possible.[6]

For more information, see *Middle East Technical University Directive For Students' Health Issues And Medical Reports*.

(24) If you have a concurrent examination scheduled for another course, then I count it as an admissible excuse provided that the instructor of the other course sends me an email through their METU email address, verifying the exam clash.

(25) For other situations (such as a family crisis, death of a relative, etc.), please contact me immediately after the exam: I may (or may not) grant you an admissible excuse.

Qualifications to Take Final Examination

(26) If you miss at least two midterm examinations without admissible excuses, you are not eligible to take the final examination: you will receive the letter grade NA.

(27) If the average of your midterm grades is below 10 out of 100,[7] then you are not eligible to take the final examination: you will receive the letter grade NA. Any missed midterm examination is exempt from the computation of this average.

(28) If you miss the final examination without an admissible excuse, you will receive the letter grade NA.

Make-up Examination

(29) The course has a single make-up examination, conducted after the final examination. The date and time of the make-up examination are announced to those students eligible to take it.

(30) The eligibility to take the make-up examination is provided in Article **(36)**.

(31) If you send me all your admissible excuses in time, I will make sure that you are counted among the students eligible to take the make-up examination. Nevertheless, please send me an email immediately after the final exam if you believe that you are eligible for the make-up examination.

(32) The make-up examination is of the same format as the final with regard to the essentials: it is a proctored written exam with the same formula sheet provided, and it encompasses the entire content of the course. However, it may contain open-ended classical questions in addition to or instead of the multiple-choice questions.

(33) You should expect the make-up to be harder than the midterms or the final as

(a) a harder make-up hedges against an intentional strategy of replacing the midterms and/or the final with the make-up for a higher grade with less effort;

(b) the make-up is conducted at the end of the semester, when the classes are over and the student has had a longer time to digest the course, so it needs to be harder; otherwise, it would be unfair to the other students.

[6] Technically, you *do not* have to send me an email; however it is much easier than the default procedure which is the following: your medical report is sent to the department secretary by the Health and Guidance Center after being approved, you regularly check with the secretary to see if it has arrived, get the document with the wet-ink signature from the secretary, and show it to all the instructors whose examination you have missed.

[7] A student who does nothing and leaves all paper empty receives $100/8 = 12.5$ points.

Semester Score and Letter Grades

(34) Letter grades are awarded based on the overall score and the attendance of the student. The effect of attendance is explained in this document; how an overall score translates into a final letter grade is determined solely by me at the end of the semester — see https://soneralbayrak.com/teaching/files/my_grading_scheme.pdf for my grading strategy in the past.

(35) For statistics of letter grades awarded in the past, you may use variations of the following link by adjusting the year, semester, and course code:
https://soneralbayrak.com/teaching/files/331_20251_infographic.png.[8]

(36) Your semester score is computed with the following algorithm:[9]

begin

Given a student who

1. has taken $0 \leq m \leq 3$ midterm and $0 \leq n \leq 1$ final examinations where each examination is graded in the range $[0, 100]$ points,
2. has the grade G_i for the i-th examination ($-\pi$ for the exams they did not take) and is assigned $\widehat{G_i} := \max(0, G_i)$,
3. has $0 \leq \alpha \leq 3$ ($0 \leq \beta \leq 1$) admissible excuses for the midterms (for the final),

define the contribution of the midterms as $G_{\text{mid}} := -\min\left(\widehat{G_1}, \widehat{G_2}, \widehat{G_3}\right) + \sum_{i=1}^{3} \widehat{G_i}$,

then their score out of 100 is computed as follows:

if $\left(m + n + \alpha + \beta \leq 2\right) \vee \left(n + \beta = 0\right) \vee \left(\sum_{i=1}^{3} \widehat{G_i} < 10\, m\right)$ **then**

the student receives the letter grade NA;

else if $m \geq 2 \ \wedge \ n = 1$ **then**

$$\boxed{\texttt{score} = \frac{1}{3}\left(G_{\text{mid}} + \widehat{G_4}\right)} \quad (\textit{final and at least 2 midterms are taken})$$

else

the student is *eligible* and *required* to take the make-up examination from which they will be assumed to receive the grade G_5:

if $G_5 = -\pi$ **then**

the student receives the letter grade NA;

else

$$\boxed{\texttt{score} = \frac{1}{3}\left(G_{\text{mid}} + \widehat{G_4} + (3 - n - \min(m, 2))\widehat{G_5}\right)} \quad (\textit{make-up replaces all missing exams})$$

end

end

end

Course Objectives, Course Goals, and Course Learning Outcomes

(37) Please see the relevant official METU Catalog webpage for these details:
https://catalog2.metu.edu.tr/course/PHYS331/2300331

[8] The correlation of letter grades awarded for the same course among different semesters is not necessarily high.

[9] The algorithm rests on two main points: (1) all midterms are equivalent, with one being treated as a bonus exam; hence, missing one exam (even with an admissible excuse) does not warrant a make-up examination; (2) although it has the same weight as the midterms, the final is treated differently in that all students should take the final (or the make-up exam if they are eligible) to pass the course.

Course Material

(38) The main textbook is *Introduction to Electrodynamics* by Griffiths (5th edition: ISBN 978-1-009-39775-9).

(39) For supplementary reading, you may also consult *Electricity and Magnetism* (Purcell & Morin), *Modern Electrodynamics* (Zangwill), and *Classical Electrodynamics* (Jackson). Note that these books are increasingly advanced, with Jackson practically being a graduate-level book.

(40) For suggested problems, please refer to
https://soneralbayrak.com/teaching/files/331332_suggestedProblems.pdf.

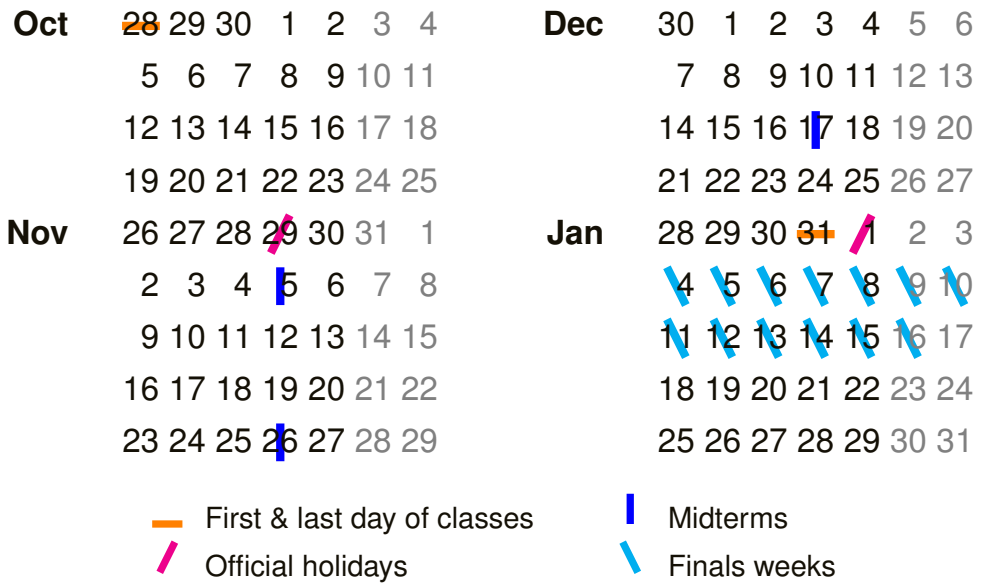
(41) For past exams, you may use variations of the following link by adjusting the year, semester, course code, and exam number:
https://soneralbayrak.com/teaching/files/331_20251_exam2.pdf.[10]

Course Timeline

(42) The weekly lecture schedule:[11]

Tuesday	13:40 – 15:30	@ P1
Wednesday	13:40 – 15:30	@ P2

(43) The semester calendar:



(44) Please be aware that November 10th, the commemoration of Atatürk, falls on a weekday this semester, although it is not an official holiday.

Miscellaneous

(45) Any topic that is not explicitly discussed in this document, such as the medium of instruction or the available letter grades, is governed by the *Middle East Technical University Rules and Regulations Governing Undergraduate Studies*.

[10] The similarity of the exams among different semesters is not necessarily high.
[11] See https://www.metu.edu.tr/location-classrooms-metu-campus for the location of classrooms.